AML Suspicious Transaction Dataset

Data card — WIE3007 Data Mining Group Project

5,000	812	206	4.12%	18	Jan–Jun 2026
transactions	accounts	suspicious	suspicious rate	columns	period

1. Purpose

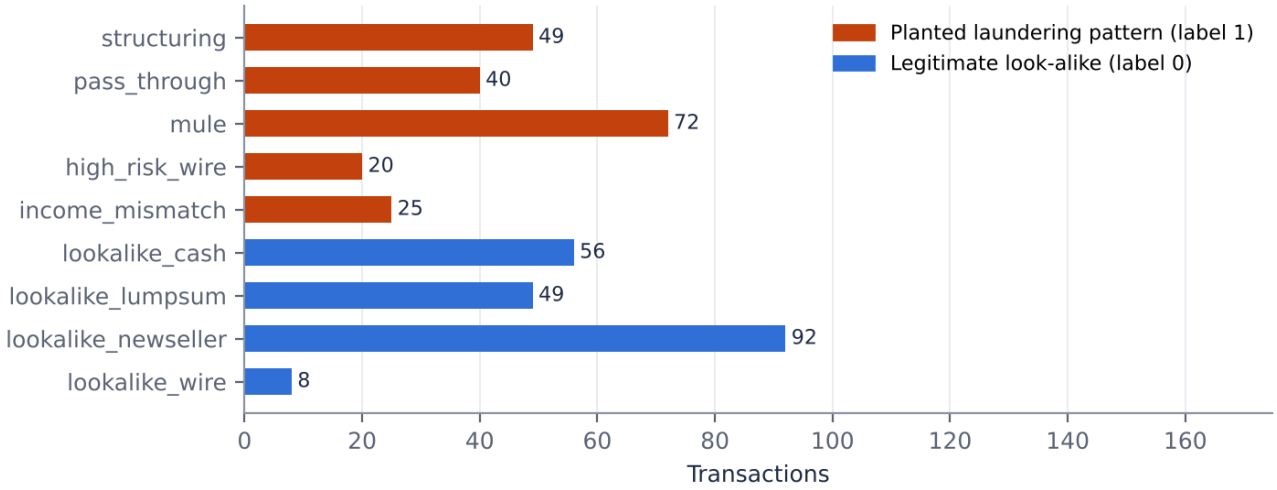
A synthetic dataset of retail and SME bank transactions in Malaysia for a **binary classification** task: predict *is_suspicious*, whether a transaction belongs to a money-laundering pattern. It combines numeric/behavioural columns with two free-text columns (*transaction_memo*, *kyc_note*) intended for LLM/SLM feature extraction, as required by the project brief (≥ 1,000 records, GenAI-created numeric and text patterns).

2. How it was generated

- **Accounts:** 812 customer accounts across four segments (retail, SME, student, high-net-worth) with occupations, declared income and an account-opening date. Most accounts are ordinary; 56 carry a laundering pattern.
- **Normal activity:** salary credits, bills, e-wallet top-ups, family transfers, supplier and customer payments, tuition and family remittances abroad — amounts drawn from log-normal distributions scaled to each customer's income.
- **Planted laundering patterns (label 1):** five typologies described in section 3.
- **Legitimate look-alikes (label 0):** genuine behaviour that resembles laundering — cash businesses depositing near the threshold, property-sale money paid out quickly, brand-new online sellers receiving many small payments, genuine trade with higher-risk countries. These stop the task from being trivially easy.
- **Behavioural columns** (*txn_count_24h*, *unique_recipients_7d*, *avg_amount_30d*, *account_age_days*) are computed from each account's own history, so they are internally consistent.
- **Text:** memo and KYC-note pools (English/Malay) were written with GenAI (Claude) and sampled with deliberate noise: ~30% of suspicious transactions have ordinary memos, ~8% of normal ones have vague memos, and ~40% of suspicious accounts have a harmless KYC note.
- **Reproducible:** *src/generate_data.py*, random seed 42. Re-running produces the identical file.

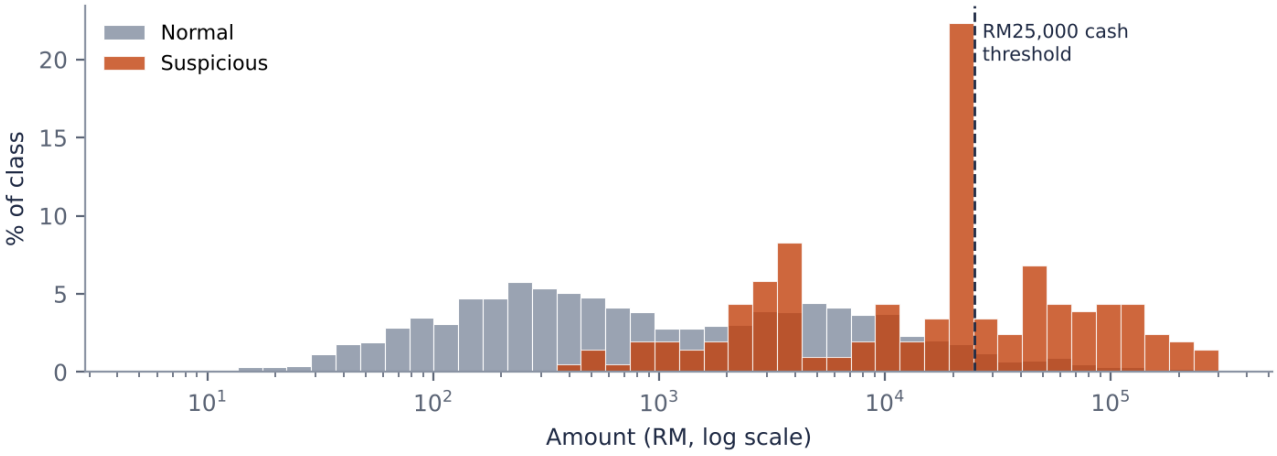
3. Planted patterns and look-alikes

Typology	Label	Rows	How it appears in the data
`structuring`	1	49	3–5 cash deposits of RM18,000–24,950 within a few days, just under the RM25,000 cash reporting threshold.
`pass_through`	1	40	Large inbound transfer/wire (RM60k–300k) moved out to 2–4 parties within hours, often at night.
`mule`	1	72	New account (opened weeks earlier) receives 3–5 small transfers from different senders, then cashes out.
`high_risk_wire`	1	20	Wires of ~RM45k (median) to high-risk jurisdictions with vague memos.
`income_mismatch`	1	25	RM15k–120k flows in accounts of low-income customers (students, part-time, unemployed).
`lookalike_cash`	0	56	Restaurants/shops banking several RM14k–24.9k takings in a week.
`lookalike_lumpsum`	0	49	Property sale, EPF withdrawal or insurance maturity paid out to developer/lawyer within a day.
`lookalike_newseller`	0	92	New online seller receives many small order payments, then pays a supplier.
`lookalike_wire`	0	8	Genuine trade payments to medium/high-risk jurisdictions with invoice references.

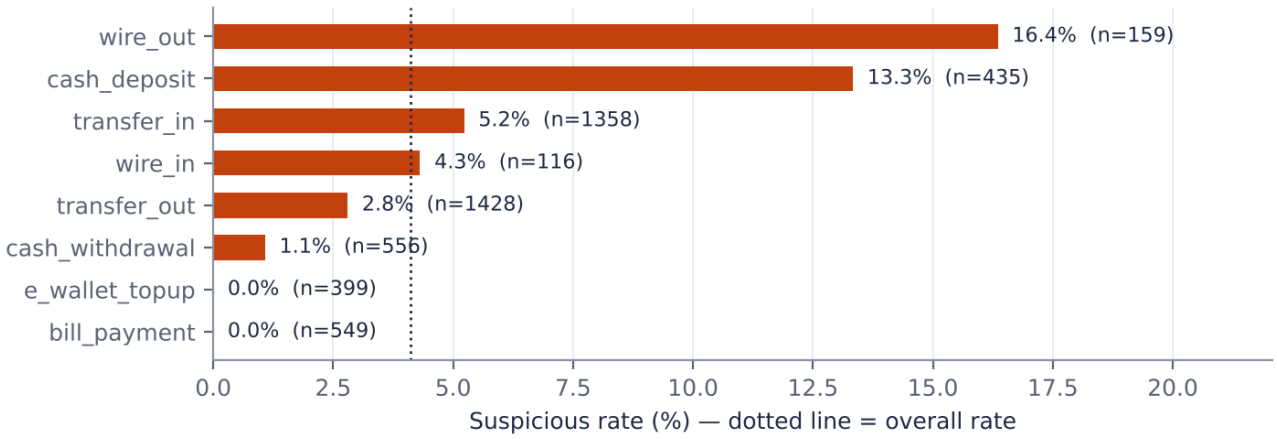


Every other row (4,589) is ordinary activity. The typology for each row is in *typology_key.csv* / the *typology_key* sheet — use it for error analysis only, never as a model feature.

4. What the data looks like



Suspicious amounts are larger on average, but the look-alikes overlap heavily — amount alone is not enough.


wire_out
cash_deposit
transfer_in
wire_in
transfer_out
cash_withdrawal
e_wallet_topup
bill_payment
16.4% (n=159)
13.3% (n=435)
5.2% (n=1358)
4.3% (n=116)
2.8% (n=1428)
1.1% (n=556)
0.0% (n=399)
0.0% (n=549)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Suspicious rate (%) — dotted line = overall rate

5. Data dictionary

Column	Type	Description	Use
`transaction_id`	text	Unique ID, in time order.	ID only
`account_id`	text	Customer account.	Group key for split
`timestamp`	datetime	Date and time (2026-01-01 to 2026-06-30).	Derive hour, weekday, night flag
`amount_myr`	numeric	Amount in RM (8.58 – 299,440).	Feature (log)
`txn_type`	category	cash_deposit, cash_withdrawal, transfer_in, transfer_out, wire_in, wire_out, e_wallet_topup, bill_payment.	Feature
`channel`	category	branch, ATM, CDM (cash deposit machine), online_banking, mobile_app.	Feature
`is_cross_border`	0/1	1 for international wires.	Feature
`dest_country_risk`	category	none (domestic), low, medium, high — FATF-style jurisdiction risk.	Feature
`counterparty_id`	text	Other party (sender or receiver).	Not a feature
`account_age_days`	integer	Days since account opening at transaction time.	Feature
`txn_count_24h`	integer	Account's transactions in the past 24h incl. this one.	Feature
`unique_recipients_7d`	integer	Distinct counterparties in the past 7 days incl. this one.	Feature
`avg_amount_30d`	numeric	Mean of the account's earlier transactions in the past 30 days; blank if none.	Feature + ratio
`occupation`	text	Declared occupation (34 values; a few blanks).	LLM → occupation_risk
`declared_monthly_income`	numeric	Declared income, RM/month (a few blanks).	Feature + ratio
`transaction_memo`	text	Customer memo, English/Malay, can be blank.	LLM → category, vagueness
`kyc_note`	text	Relationship-manager note, one per account.	LLM → kyc_risk_tone
`is_suspicious`	0/1	TARGET — 206 positives (4.12%).	Target

6. Data quality notes (for EDA)

- **avg_amount_30d** is blank in 1,994 rows (40%) because the account had no earlier transaction in the previous 30 days. This is expected — impute (e.g. with the transaction amount, so the ratio = 1) and/or add a 'no recent history' flag.
- **occupation** is blank in 63 rows and **declared_monthly_income** in 93 rows (a few accounts with incomplete onboarding data) — left in deliberately so EDA and preprocessing have something to handle.
- **transaction_memo** is blank in 194 rows — real customers often skip the memo. Treat blank as its own category.
- Amounts are heavily right-skewed: use a log transform for Logistic Regression, MLP and k-NN.
- No duplicate transaction IDs; all amounts are positive; timestamps are within the stated period.

7. Rules for the modelling team

- **Split by account:** *GroupShuffleSplit(test_size=0.2, random_state=42)* on *account_id*.
- **Never use** typology, transaction_id or counterparty_id as features.
- **LLM inputs** are only the text columns (memo, KYC note, occupation) — never the label.
- **Metrics:** Recall, Precision, F1 and PR-AUC on the suspicious class, plus ROC-AUC. Accuracy is misleading at ~4% positives.
- **Imbalance:** class weights, or SMOTE applied to the training set only.

8. Sample rows (15 of 5,000; full data in the Excel/CSV file)

txn_id	account	timestamp	amount (RM)	type	channel	dest risk	acct age	cnt 24h	recip 7d	occupation	memo	kyc_note	label
TXN000019	ACC00363	2026-01-01 18:08:23	666.44	transfer_in	mobile_app	none	4591	1	1	Bank Officer	insurance claim	Customer updated address and phone number; supporting documents verified.	0
TXN000061	ACC00101	2026-01-03 10:41:13	242.82	transfer_out	mobile_app	none	1165	1	1	Homemaker	PTPTN	Customer verified in branch with MyKad and payslip. No adverse findings.	0
TXN000288	ACC00450	2026-01-10 12:50:33	287.70	bill_payment	mobile_app	none	4179	1	1	Homemaker	Unifi bill	Customer is a regular walk-in, friendly and cooperative during periodic review.	0
TXN001039	ACC00587	2026-02-08 04:35:42	2,890.95	transfer_in	mobile_app	none	3844	1	1	Money Changer	Payment for INV 73505	Periodic review completed. No change in risk profile.	0
TXN001721	ACC00768	2026-03-05 00:40:54	3,648.50	transfer_in	mobile_app	none	16	3	4	Unemployed	refund shopee	Customer requested to split deposit into smaller amounts at the counter.	1
TXN002681	ACC00798	2026-04-06 17:41:43	2,658.27	transfer_in	mobile_app	none	38	1	4	Student	preorder	Customer updated address and phone number; supporting documents verified.	0
TXN002878	ACC00776	2026-04-13 18:00:00	6,984.16	transfer_out	mobile_app	none	29	1	5	—	wedding gift	Customer appeared to be accompanied and instructed by another person at the branch.	1
TXN003403	ACC00625	2026-05-01 11:00:03	6,345.08	transfer_out	mobile_app	none	719	2	2	Hardware Shop Owner	Staff salary May	Periodic review completed. No change in risk profile.	0
TXN003643	ACC00746	2026-05-11 07:51:06	21,850.17	cash_deposit	branch	none	2027	3	3	Car Dealer	USDT	Address verification still pending; mail returned undelivered.	1
TXN003994	ACC00760	2026-05-24 08:13:18	256,622.25	transfer_in	online_banking	none	2702	1	1	Money Changer	-	Business registered with SSM; turnover broadly consistent with bank statements provided.	1
TXN004256	ACC00289	2026-06-02 14:09:41	97,785.69	transfer_in	online_banking	none	643	1	2	Civil Servant	Fixed deposit uplift	Customer is a regular walk-in, friendly and cooperative during periodic review.	0
TXN004457	ACC00758	2026-06-10 23:36:37	48,249.38	transfer_out	online_banking	none	1836	4	4	Bank Officer	help friend	Joint relationship with spouse; usage mostly household expenses.	1
TXN004503	ACC00596	2026-06-12 17:36:54	10,204.52	transfer_in	mobile_app	none	2996	1	1	Car Dealer	INV-67300 payment	Customer verified in branch with MyKad and payslip. No adverse findings.	0
TXN004643	ACC00149	2026-06-17 17:02:14	60,942.22	transfer_in	online_banking	none	302	1	1	Accountant	EPF withdrawal	Online-only customer, eKYC passed, low and regular activity.	0
TXN004669	ACC00753	2026-06-18 21:58:29	51,516.50	wire_out	online_banking	high	2007	2	2	Jewellery Shop Owner	help friend	Address verification still pending; mail returned undelivered.	1

Orange label cells = suspicious. The sample deliberately mixes suspicious, look-alike and ordinary rows.